Tidal Energy Harvesting Efficiency in Coastal Turbine Arrays

Marie Curie, Nikola Tesla

Abstract

This paper studies the efficiency of underwater turbines that convert tidal currents into electricity across shallow coastal shelves. We measure power output against tidal flow velocity and blade geometry.

1. Introduction

Coastal tides are a predictable renewable resource. Submerged turbine arrays capture kinetic energy from the moving water column as tides ebb and flood twice daily along the shoreline.

2. Methodology

We deploy a scaled turbine array in a tidal channel and log rotor torque against measured current speed over a full lunar cycle, then compute the hydrodynamic capacity factor of each blade profile.

3. Conclusion

Optimally spaced coastal turbines sustain high capacity factors and make tidal arrays a viable baseload marine energy source.